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Department of Engineering

Session 2026-27

PROMPT ENGINEERING

(Course Code: BCO610A)

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Semester - 3

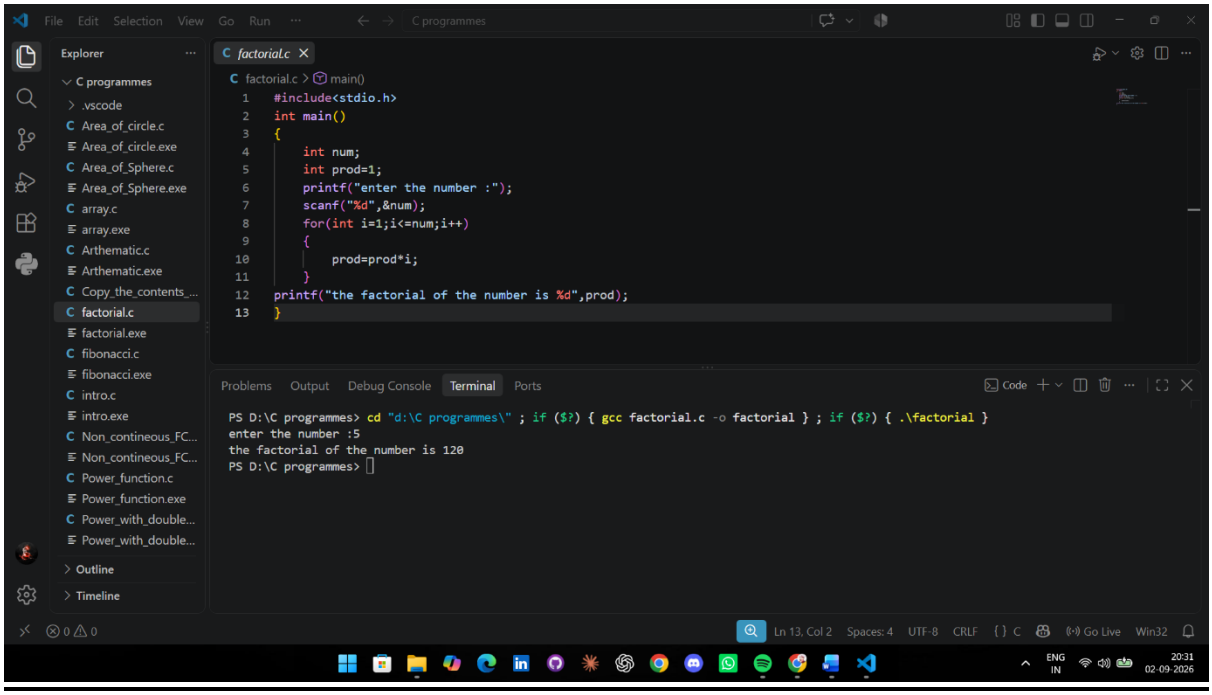
Submitted To:

Mr. Ajay Sharma

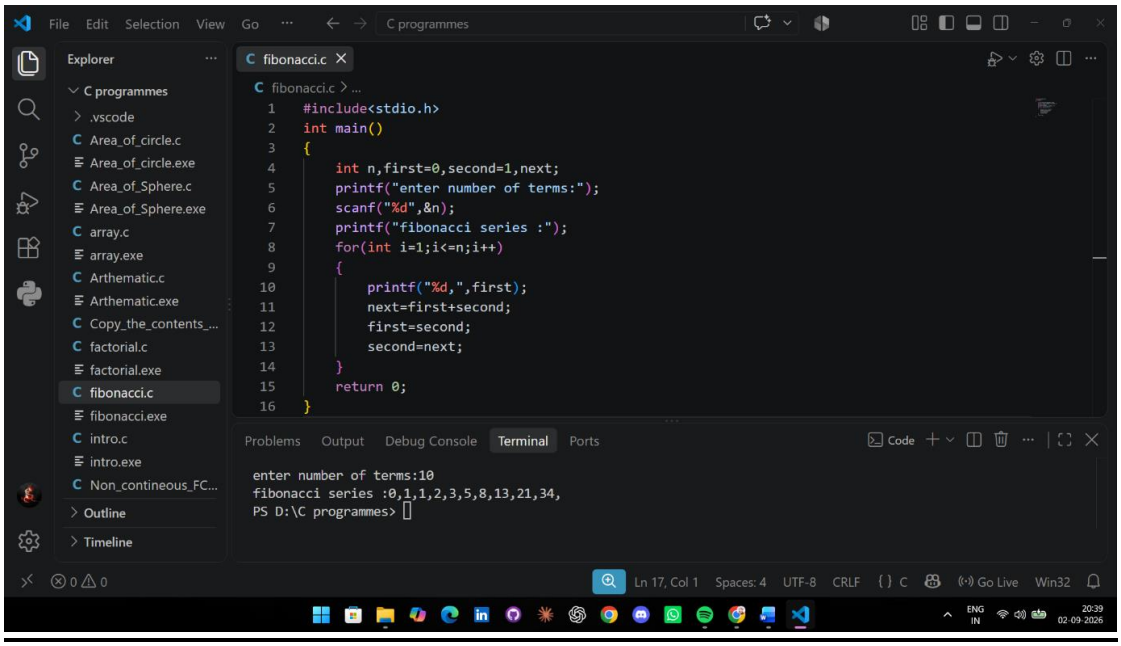
Prompt Engineering

C – Programmes

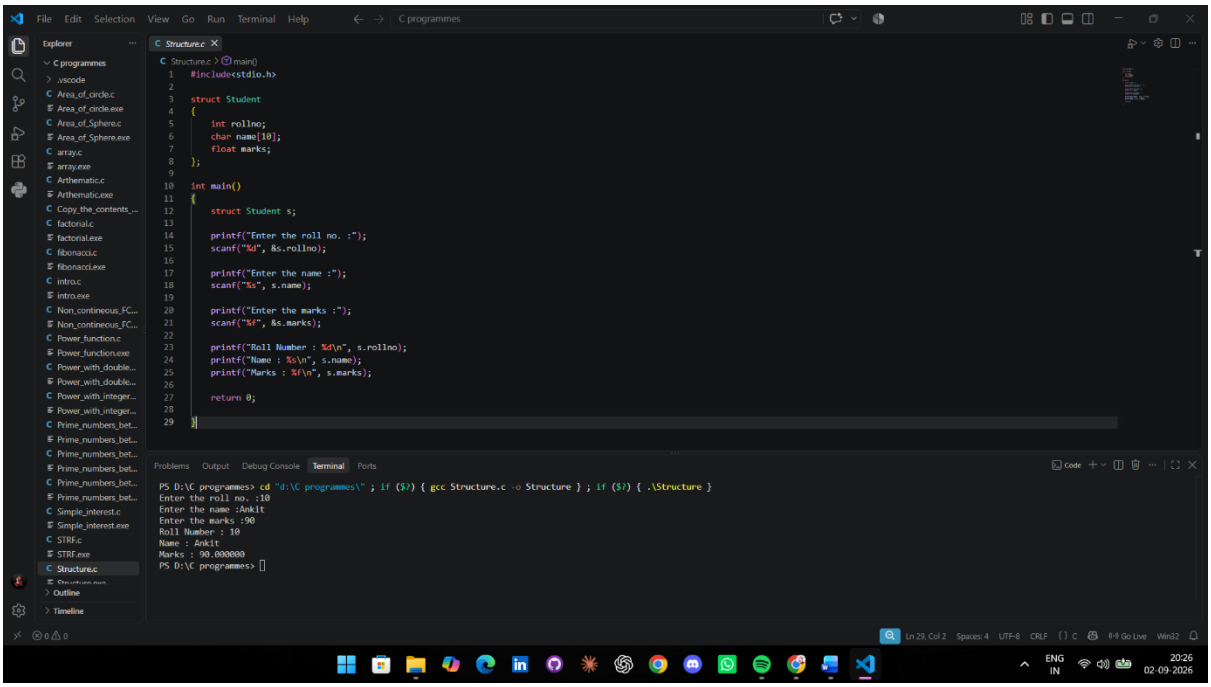
1. Factorial :



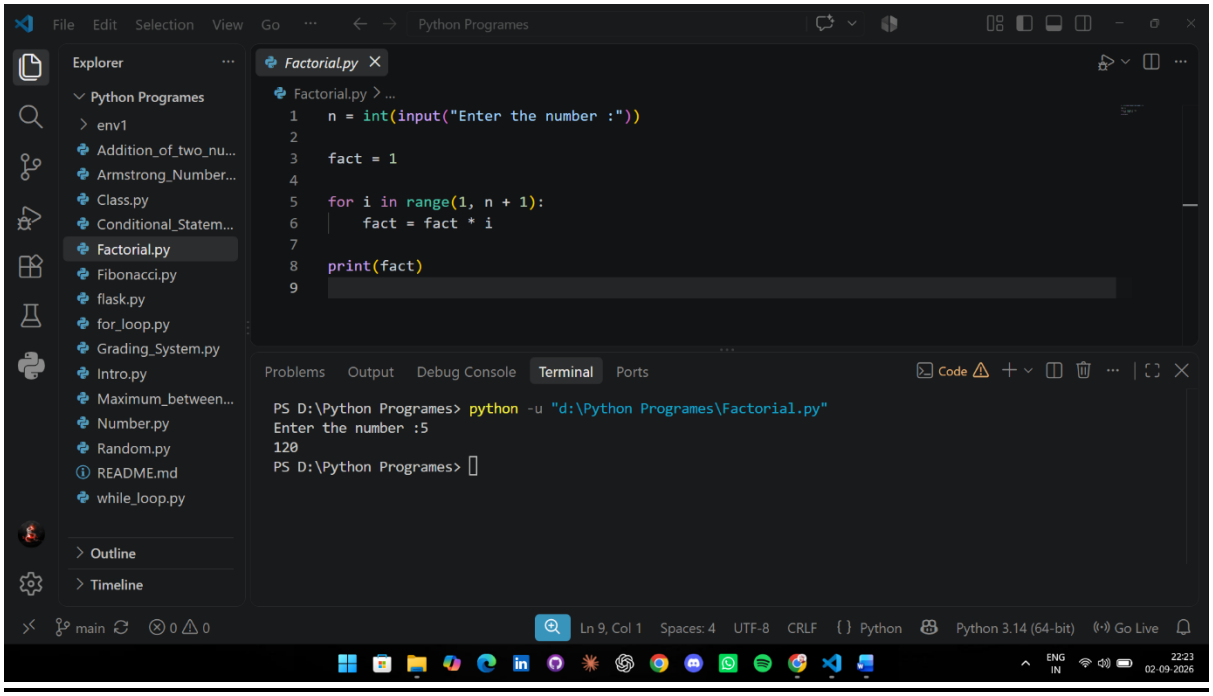
2. Fibonacci :



3. Structure :

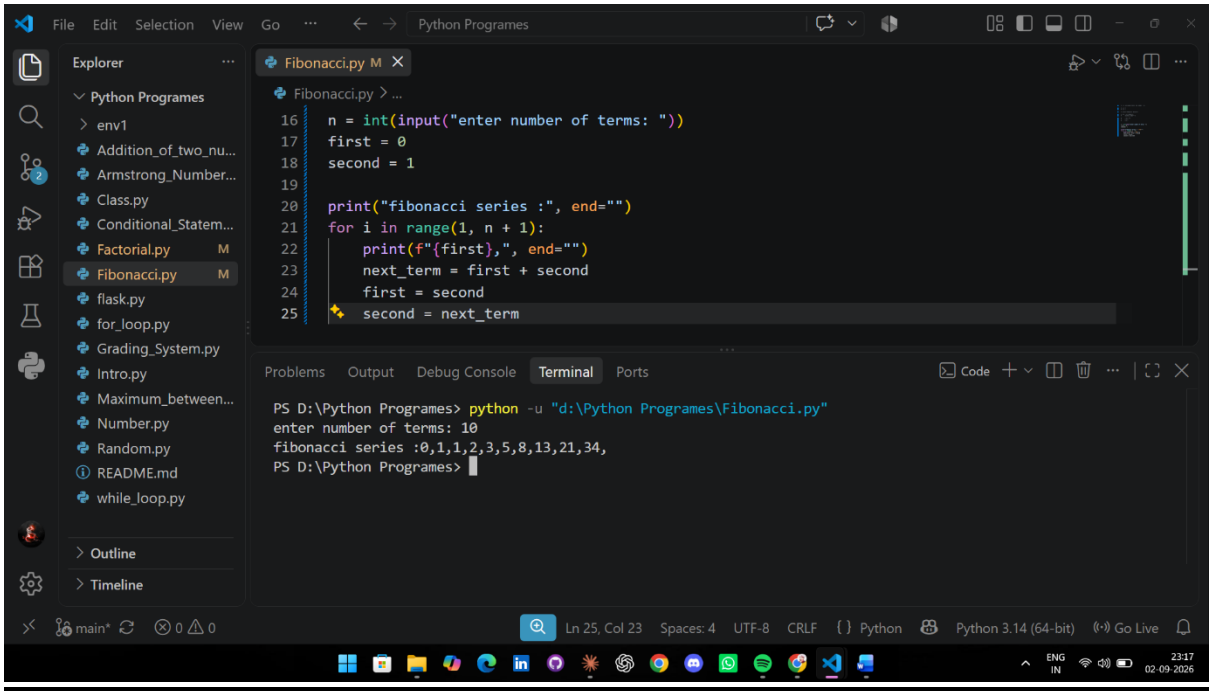


Factorial Programme in Python without AI

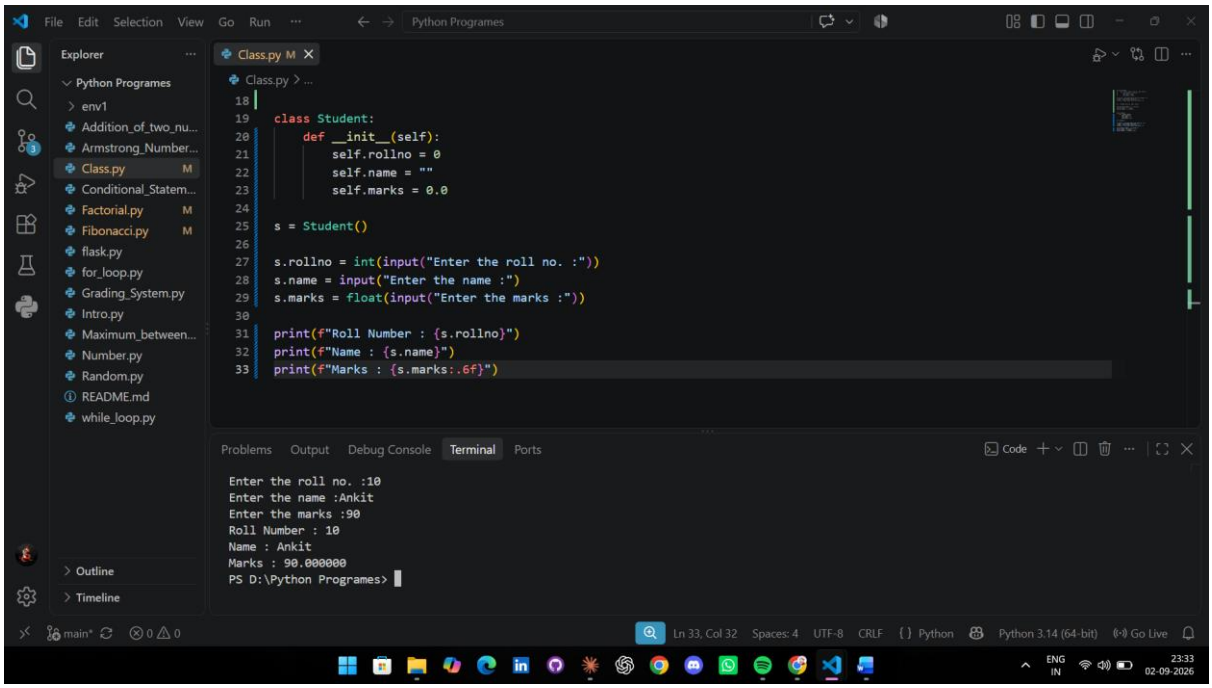


AI – Assisted Translation

1. Fibonacci :



2. Class :



Mismatches and Fixes

1. Fibonacci — Mismatches and Fixes :

No.	Mismatch / Issue	C	Python Fix
1	Variable declaration	`int a, b, c;`	Python does not require datatype declarations: `a, b, c = 0, 1, 0`
2	Input method	`scanf("%d", &n);`	Use `n = int(input("Enter number of terms: "))`
3	Output method	`printf("%d ", a);`	Use `print(a, end=" ")`
4	Loop syntax	`for(i=0; i<n; i++)`	Use `for i in range(n):`
5	Increment operator	`a++`	Python does not have `++`; use `a += 1`
6	Temporary value	`c = a + b;`	Same logic: `c = a + b`
7	Updating variables	`a = b; b = c;`	Same logic can be used: `a, b = b, c`
8	Datatype handling	C requires `int` declaration	Python automatically handles integer variables

2. Structure — Mismatches and Fixes :

2. Structure — Mismatches and Fixes

No.	Mismatch / Issue	C	Python Fix
1	Structure declaration	`struct Student { ... };`	Python can use a `class` or dictionary
2	Variable declaration	`int roll;`	Python: `roll = 101`
3	String declaration	`char name[50];`	Python: `name = "Ankit"`
4	Accessing members	`s.roll`	Python class also uses `s.roll`
5	Input	`scanf("%d", &s.roll);`	`s.roll = int(input())`
6	String input	`scanf("%s", s.name);`	`s.name = input()`
7	Structure variable	`struct Student s;`	`s = Student()` when using a class
8	Datatypes	C requires explicit datatypes	Python determines the datatype automatically

Review of the hand written Factorial program

* input() takes the number from the user.
* int() converts the input into an integer.
* fact = 1 initializes the factorial.
* range(1, n + 1) generates numbers from 1 to n.
* fact = fact * i multiplies each number into fact.
* print(fact) displays the final factorial.

Verdict of the hand written Factorial program

The handwritten Python program is correct. It successfully calculates the factorial of a given number using a for loop. There are no major syntax or logical errors in the program.

8-Row Cheat Sheet

No.	Concept	C	Python
1	Output	`printf("Hello");`	`print("Hello")`
2	Input	`scanf("%d", &n);`	`n = int(input())`
3	Variable Declaration	`int n = 10;`	`n = 10`
4	If-Else	`if(n > 0) { ... }`	`if n > 0:`
5	For Loop	`for(i=0; i<n; i++)`	`for i in range(n):`
6	While Loop	`while(n > 0) { ... }`	`while n > 0:`
7	Array / List	`int a[5] = {1,2,3,4,5};`	`a = [1,2,3,4,5]`
8	Function	`int add(int a,int b) { return a+b; }`	`def add(a,b): return a+b`

Personal Reflection

Personal Reflection

Through this assignment, I learned how to translate C programs into Python and understand the differences between the two languages. I learned that Python has simpler syntax and does not require explicit variable declarations like C. While translating the **Fibonacci** and **Structure** programs, I understood the differences in loops, input/output, arrays, and data structures.

I also learned that AI-generated code should not be accepted blindly. Sometimes the logic or syntax may

need to be checked and corrected. By running the programs and comparing their outputs, I learned how to identify mismatches and fix them.
Reviewing the handwritten **factorial program** helped me understand how to check a program for syntax and logical errors. Overall, this assignment improved my programming skills and gave me a better understanding of how the same problem can be solved in both C and Python.